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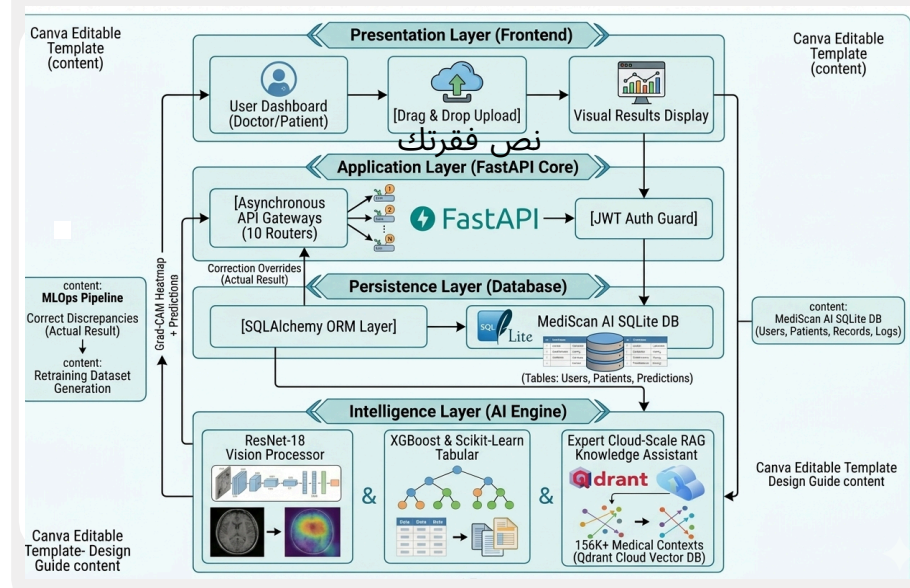
Undersupervision:

Dr: Mussad Wajih

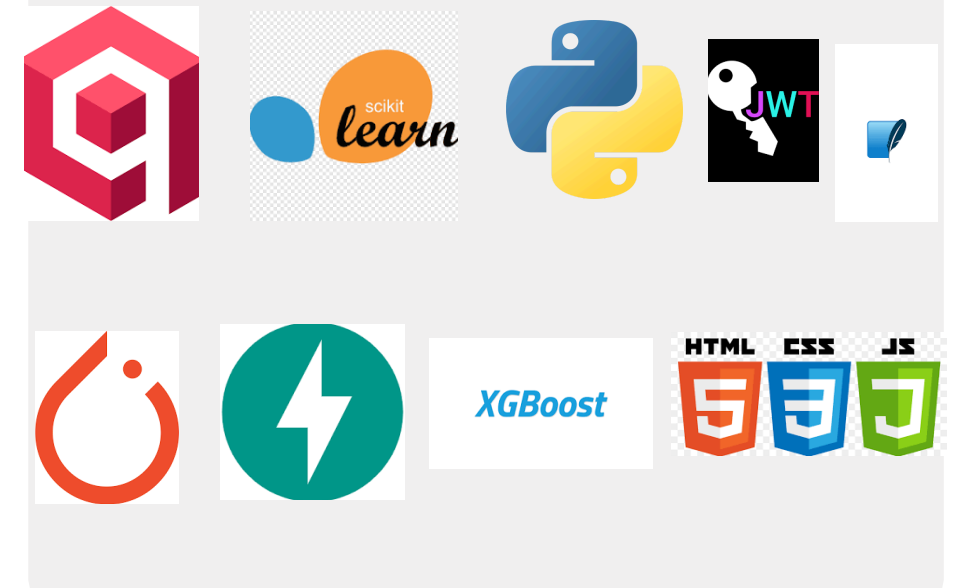
Abstract

- MediScan AI is a medical platform that automates multi-disease screening using DL and ML pipelines
- It processes multi-modal data including medical imaging (MRI, CT, X-ray, Fundus) and structured clinical biomarkers
- Integrates Grad-CAM for explainable AI, an advanced Cloud RAG engine, and a Human-in-the-Loop pipeline for real-time history tracking

System Architecture



Technologies



Problem Statement

Diagnostic Bottlenecks: Shortage of radiologists causes severe delays in scanning MRI/CT/X-rays, risking late disease discovery.

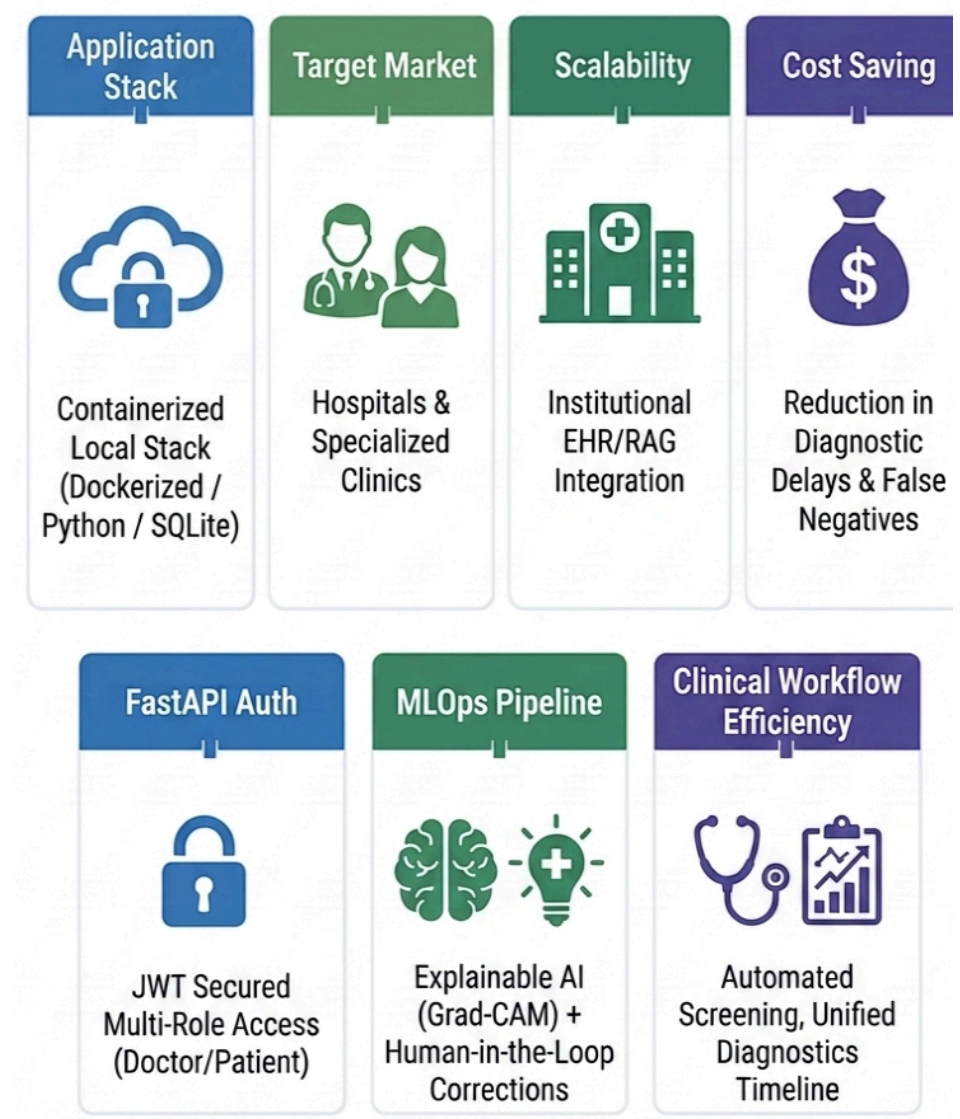
- "Black-Box" AI Dilemma: Traditional deep models lack visual interpretability, preventing clinical trust and adoption.
- Stateless & Isolated Systems: Existing tools lack database persistence, failing to track a patient's longitudinal history

Results & Benefits

- Robust Multi-Modal Accuracies (Pure Test Sets):
- Brain Stroke (CT): 97.0% Accuracy (6,650 slices total dataset)
- Diabetic Retinopathy: 98.0% Accuracy (3,662 balanced images)
- Chest X-Ray: 88.1% Accuracy with 99.7% Recall for high patient safety
- Brain Tumor (MRI): 81.2% Accuracy (92.2% AUC multi-class ResNet18)
- CKD Staging (Tabular): 99.0% Accuracy across 6 clinical stages.
- Diabetes Predictor: 96.4% Accuracy via optimized XGBoost pipelines

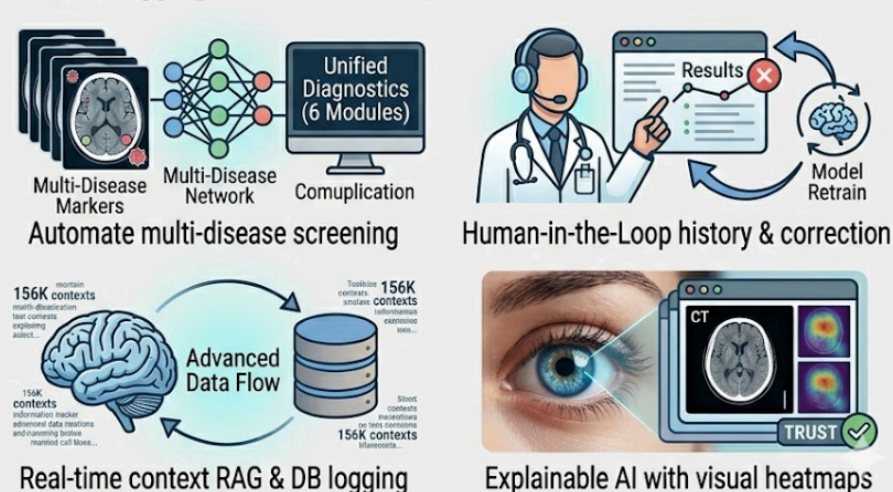


Business Model



Objectives

- Multi-Modal Screening: Deliver 6 diagnostic modules (imaging + tabular) via a unified platform
- Explainable AI & Trust: Integrate Grad-CAM visual heatmaps to eliminate "black-box" clinical risks
- Data Persistence & MLOps: Enforce secure SQLite history tracking with Human-in-the-Loop model retraining.



Conclusion

- Production-Ready Paradigm: MediScan AI successfully transitions static healthcare models into a secure, role-based desktop ecosystem leveraging robust local persistence.
- Bridging the Clinical Gap: By combining multi-modal screening, visual interpretability (Grad-CAM), and MLOps continuous loop learning, the platform effectively builds institutional trust and eliminates diagnostics bottlenecks.